

The empirical recurrence for OEIS A265930, recovered from its tail

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Abstract

OEIS A265930 records “Empirical recurrence of order 55” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 91$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 60$. Its last coefficient is $c_{55} = 89060441849856 \neq 0$, so no further tail satisfies anything shorter; any order-55 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A265930 is “Number of $2 \times n$ 0..4 arrays with the absolute differences of each element with its with horizontal and antidiagonal neighbors unique.”. It has offset 1 and begins

25, 340, 1740, 9016, 44916, 223788, 1119424, 5621396, ...

The entry states:

Empirical recurrence of order 55 (see link above)

As of the “Last modified” line on the live entry (Dec 19 2015 09:51:01, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 91$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 91$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 55: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 55.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 182$ exact terms from $n = 5$ on, it returns order 55 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{55} c_i a(n-i),$$

c_1	1	c_{15}	5117778	c_{29}	−7253357632	c_{43}	−24855844487168
c_2	6	c_{16}	20675755	c_{30}	−110394682368	c_{44}	11877035802624
c_3	26	c_{17}	27672925	c_{31}	−307538955520	c_{45}	23875186327552
c_4	203	c_{18}	17323544	c_{32}	−523965277184	c_{46}	88667896938496
c_5	81	c_{19}	−55041404	c_{33}	−473252193792	c_{47}	61481735421952
c_6	718	c_{20}	−356154872	c_{34}	−273106449152	c_{48}	35186788007936
c_7	1554	c_{21}	−843857072	c_{35}	1314167528192	c_{49}	−56445533945856
c_8	−14096	c_{22}	−1761657016	c_{36}	2304392810496	c_{50}	−112841675767808
c_9	−33712	c_{23}	−1281858176	c_{37}	4474638474240	c_{51}	−97298189123584
c_{10}	−54298	c_{24}	817391216	c_{38}	3693951791104	c_{52}	−88235808129024
c_{11}	−136570	c_{25}	7763815952	c_{39}	977921925120	c_{53}	56899726737408
c_{12}	−176837	c_{26}	24442345808	c_{40}	−6334276763648	c_{54}	44530220924928
c_{13}	497609	c_{27}	35517084944	c_{41}	−19377870897152	c_{55}	89060441849856
c_{14}	1846654	c_{28}	48622059520	c_{42}	−18611474989056		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 60$, and it is the recurrence OEIS A265930 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{55} - c_1 x^{54} - \dots - c_{55}$. Its constant term is $-c_{55} = -89060441849856$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 55 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 55, so $\deg q = 55$, and it is monic. It holds from some index t on. From

$\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 55, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 55, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 89060441849856.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-55 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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